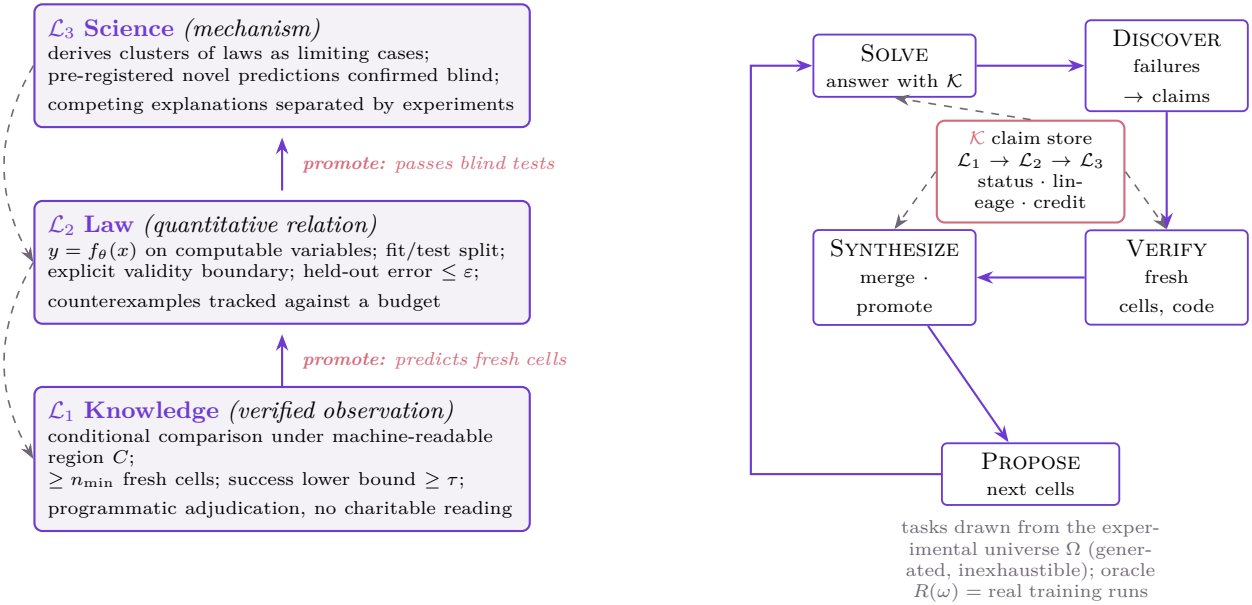
$\mathcal{L}_3$ Science (mechanism)
derives clusters of laws as limiting cases;
pre-registered novel predictions confirmed blind;
competing explanations separated by experiments
promote: passes blind tests
$\mathcal{L}_2$ Law (quantitative relation)
$y = f_\theta(x)$ on computable variables; fit/test split;
explicit validity boundary; held-out error $\leq \varepsilon$;
counterexamples tracked against a budget
promote: predicts fresh cells
$\mathcal{L}_1$ Knowledge (verified observation)
conditional comparison under machine-readable region C;
$\geq n_{\min}$ fresh cells; success lower bound $\geq \tau$;
programmatic adjudication, no charitable reading
SOLVE
answer with $\mathcal{K}$
DISCOVER
failures
→ claims
$\mathcal{K}$ claim store
$\mathcal{L}_1 \to \mathcal{L}_2 \to \mathcal{L}_3$
status · lineage · credit
SYNTHESIZE
merge · promote
VERIFY
fresh cells, code
PROPOSE
next cells
tasks drawn from the experimental universe Ω (generated, inexhaustible); oracle $R(\omega)$ = real training runs